



DSA

Module 2

Math

Author
Srinivas Dande



My **JAVA
LEARNING
CENTER**



2.1. Programs on Math

1) Lab1.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

// 1. Swap 2 Numbers.

public class Lab1 {

    public static void main(String[] args) {
        int a=10;
        int b=20;

        System.out.println(a+"\t"+b);

        int temp= a;
        a=b;
        b=temp;

        System.out.println(a+"\t"+b);

    }

}

// Time Complexity - O(1)
//Space Complexity - O(1)
```



2) Lab2.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//2. Swap 2 Numbers without 3rd Variable

public class Lab2 {

    public static void main(String[] args) {
        int a=10;
        int b=20;

        System.out.println(a+"\t"+b);

        a=a+b;
        b=a-b;
        a=a-b;

        System.out.println(a+"\t"+b);
    }
}

// Time Complexity - O(1)
// Space Complexity - O(1)
//This Logic Works only for Numbers.
```



3) Lab3.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//3. Swap 2 Strings

public class Lab3 {

    public static void main(String[] args) {
        String str1="Hello";
        String str2="JLC";

        System.out.println(str1+"\t"+str2);

        String temp= str1;
        str1=str2;
        str2=temp;

        System.out.println(str1+"\t"+str2);

    }

}

// Time Complexity - O(1)
//Space Complexity - O(1)
```



4) Lab4.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

//4. Find Min and max of 2 Numbers

```
public class Lab4 {  
  
    public static void main(String[] args) {  
        int a=10;  
        int b=20;  
  
        int min = (a<b) ? a:b;  
        System.out.println(min);  
  
        int max=0;  
  
        if(a<b)  
            max=a;  
        else  
            max=b;  
  
        System.out.println(max);  
    }  
}  
  
// Time Complexity - O(1)  
//Space Complexity - O(1)
```



5) Lab5.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//5. Find max of 3 Numbers

class Lab5 {
    public static void main(String as[]) {
        int a=10;
        int b=20;
        int c=15;

        int max=(a>b) ? ( (a>c) ? a : c) : ((b>c) ? b : c);
        System.out.println(max);

        if(a>b && a>c)
            max=a;
        else if(b>a && b>c)
            max = b;
        else
            max=c;

        System.out.println("Max value is "+max);
    }
}

// Time Complexity - O(1)
//Space Complexity - O(1)
```



6) Lab6.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//6. Print the Numbers from 1 to N

public class Lab6 {

    public static void main(String[] args) {
        int n = 25;

        for (int i = 1; i <= n; i++) {
            System.out.println(i);
        }

    }

}

// Time Complexity - O(n)
//Space Complexity - O(1)
```



7) Lab7.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//7. Print the Even Numbers from 1 to N

public class Lab7 {

    public static void main(String[] args) {
        int n = 6;

        for (int i = 1; i <= n; i++) {
            if (i % 2 == 0)
                System.out.println(i);
        }

    }

}

// Time Complexity - O(n)
//Space Complexity - O(1)
```



8) Lab8.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//8. Find the Sum of Numbers from 1 to N

public class Lab8 {

    public static void main(String[] args) {
        int n = 6;

        int sum=0;
        for (int i = 1; i <= n; i++) {
            sum = sum+i;
        }

        System.out.println("Sum : "+sum);
    }
}

// Time Complexity - O(n)
//Space Complexity - O(1)
```



9) Lab9.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

```
//9. Find the Sum of Numbers from 1 to N
```

```
public class Lab9 {  
  
    public static void main(String[] args) {  
        int n =5;  
  
        int sum=n * (n+1)/2;  
        System.out.println("Sum : "+sum);  
  
    }  
  
}
```

```
// Time Complexity - O(1)
```

```
//Space Complexity - O(1)
```



10) Lab10.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

```
//10. Find the Sum of Squares of Numbers from 1 to N
```

```
public class Lab10 {  
  
    public static void main(String[] args) {  
        int n = 3;  
  
        int sum=0;  
        for (int i = 1; i <= n; i++) {  
            sum = sum+i*i;  
        }  
  
        System.out.println("Sum : "+sum);  
    }  
}  
  
// Time Complexity - O(n)  
//Space Complexity - O(1)
```



11) Lab11.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

```
//11. Find the Sum of Cubes of Numbers from 1 to N
```

```
public class Lab11 {  
  
    public static void main(String[] args) {  
        int n = 5;  
  
        long sum=0;  
        for (int i = 1; i <= n; i++) {  
            sum = sum+i*i*i;  
        }  
  
        System.out.println("Sum : "+sum);  
    }  
}  
  
// Time Complexity - O(n)  
//Space Complexity - O(1)
```



12) Lab12.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//12. Print the Divisibles of 6 from 1 to N

public class Lab12 {
    public static void main(String[] args) {
        int n = 25;
        for (int i = 1; i <= n; i++) {
            if (i % 6 == 0)
                System.out.println(i);
        }
    }
}

// Time Complexity - O(n)
//Space Complexity - O(1)
```



13) Lab13.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//13. Print the Divisibles of given Number

public class Lab13 {
    public static void main(String[] args) {
        int n = 10;

        for (int i = 1; i <= n/2; i++) {
            if (n % i == 0)
                System.out.println(i);
        }
        System.out.println(n);
    }
}

// Time Complexity - O(n/2)
//Space Complexity - O(1)
```



14) Lab14.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
**/
```

```
//14. Find the Sum of Divisibles of given Number(exclude N)
```

```
public class Lab14 {  
    public static void main(String[] args) {  
        int n = 28;  
  
        int sum = 0;  
        for (int i = 1; i <= n/2; i++) {  
            if (n % i == 0)  
                sum +=i;  
        }  
        System.out.println("Sum : "+ sum);  
    }  
}
```

```
// Time Complexity - O(n/2)
```

```
//Space Complexity - O(1)
```



15) Lab15.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

//15. Check whether Given Number is Prime or Not

```
public class Lab15 {  
  
    public static boolean isPrime(int n) {  
        int count = 0;  
        for (int i = 2; i <= n / 2; i++) {  
            if (n % i == 0)  
                count++;  
        }  
  
        if (count == 0)  
            return true;  
        else  
            return false;  
    }  
  
    public static void main(String[] args) {  
        int n = 10;  
        boolean flag = isPrime(n);  
        System.out.println(flag);  
    }  
}
```

// Time Complexity - $O(n/2)$

//Space Complexity - $O(1)$



16) Lab16.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//16. Check whether Given Number is Prime or Not
// Optimized

public class Lab16 {

    // Time Complexity - O(log n)
    // Time Complexity - O(n/2)
    public static boolean isPrime(int n) {
        int count = 0;
        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                count++;
                break;
            }
        }

        if (count == 0)
            return true;
        else
            return false;
    }

    public static void main(String[] args) {
        int n = 100;
        boolean flag = isPrime(n);
        System.out.println(flag);
    }
}
```



17) Lab17.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//17. Check whether Given Number is Prime or Not
// More Optimized

public class Lab17 {

    // Time Complexity -  $O(\log n)$ 
    // Time Complexity -  $O(n/2)$ 
    public static boolean isPrime(int n) {

        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                return false;
            }
        }
        return true;
    }

    public static void main(String[] args) {
        int n = 10;
        boolean flag = isPrime(n);
        System.out.println(flag);
    }
}
```



18) Lab18.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//18. Print Prime Numbers from 1 to N

public class Lab18 {

    public static boolean isPrime(int n) {

        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                return false;
            }
        }
        return true;
    }

    public static void main(String[] args) {
        int n = 50;
        for(int i=1;i<=n;i++) {
            if(isPrime(i))
                System.out.println(i);
        }
    }

}

// Time Complexity - O(n/2)
//Space Complexity - O(1)
```



19) Lab19.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

```
//14. Count the Divisibles of given Number
```

```
public class Lab19 {  
  
    public static void main(String[] args) {  
        int n = 10;  
  
        int count = 2;  
        for (int i = 2; i <= n/2; i++) {  
            if (n % i == 0)  
                count++;  
        }  
  
        System.out.println("Count : "+ count);  
    }  
}  
  
// Time Complexity - O(n/2)  
//Space Complexity - O(1)
```



20) Lab20.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

//20. Check whether given number is Perfect Number or Not

```
public class Lab20 {  
  
    public static boolean isPerfect(int n) {  
        int sum = 0;  
        for (int i = 1; i <= n / 2; i++) {  
            if (n % i == 0)  
                sum += i;  
        }  
  
        if (sum == n)  
            return true;  
        else  
            return false;  
    }  
  
    public static void main(String[] args) {  
        int n = 60;  
        boolean flag = isPerfect(n);  
        System.out.println(flag);  
    }  
}
```

// Time Complexity - $O(n/2)$

//Space Complexity - $O(1)$



21) Lab21.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//21. Print Perfect Numbers from 1 to N

public class Lab21 {

    public static boolean isPerfect(int n) {
        int sum = 0;
        for (int i = 1; i <= n / 2; i++) {
            if (n % i == 0)
                sum += i;
        }

        if (sum == n)
            return true;
        else
            return false;
    }

    public static void main(String[] args) {
        int n = 10000;
        for(int i=1;i<=n;i++) {
            if(isPerfect(i))
                System.out.println(i);
        }
    }

    // Time Complexity -  $O(n * n/2)$  /  $O(n^2)$ 
    //Space Complexity -  $O(1)$ 
```



22) Lab22.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//22. Print the Digits of Given Number

public class Lab22 {

    public static void main(String[] args) {
        int n = 12345;

        while(n!=0) {
            int digit = n%10;
            System.out.println(digit);
            n= n/10;
        }
    }
}

// Time Complexity - O(log n)
//Space Complexity - O(1)
```



23) Lab23java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

```
//23. Count the Digits of Given Number
```

```
public class Lab23 {  
  
    public static void main(String[] args) {  
        int n = 12345;  
  
        int count=0;  
        while(n!=0) {  
            //int digit = n%10;  
            count++;  
            n= n/10;  
        }  
        System.out.println(count);  
    }  
}
```

```
// Time Complexity - O(log n)
```

```
//Space Complexity - O(1)
```



24) Lab24.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//24. Sum the Digits of Given Number

public class Lab24 {

    public static void main(String[] args) {
        int n = 123;

        int sum=0;
        while(n!=0) {
            int digit = n%10;
            sum = sum +digit;
            n= n/10;
        }
        System.out.println(sum);
    }

}

// Time Complexity - O(log n)
//Space Complexity - O(1)
```



25) Lab25.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

```
//25. Sum the Squares of Digits of Given Number
```

```
public class Lab25 {  
  
    public static void main(String[] args) {  
        int n = 123;  
  
        int sum=0;  
        while(n!=0) {  
            int digit = n%10;  
            sum = sum +digit * digit;  
            n= n/10;  
        }  
        System.out.println(sum);  
    }  
}
```

```
// Time Complexity -  $O(\log n)$ 
```

```
//Space Complexity -  $O(1)$ 
```



26) Lab26.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//26. Sum the Cubes of Digits of Given Number

public class Lab26 {

    public static void main(String[] args) {
        int n = 123;

        int sum=0;
        while(n!=0) {
            int digit = n%10;
            sum = sum +digit * digit * digit;
            n= n/10;
        }
        System.out.println(sum);
    }

}

// Time Complexity - O(log n)
//Space Complexity - O(1)
```



27) Lab27.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

//27. Check whether given number is Arm Strong Number or Not

```
public class Lab27 {  
    public static boolean isArmStrong(int n) {  
  
        int nCopy=n;  
        int sum = 0;  
        while (n != 0) {  
            int digit = n % 10;  
            sum = sum + digit * digit * digit;  
            n = n / 10;  
        }  
  
        if (sum ==nCopy)  
            return true;  
        else  
            return false;  
    }  
  
    public static void main(String[] args) {  
        int n = 372;  
        boolean flag = isArmStrong(n);  
        System.out.println(flag);  
    }  
}
```

// Time Complexity - $O(\log n)$

//Space Complexity - $O(1)$



28) Lab28.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//28. Print Arm Strong Numbers from 1 to N

public class Lab28 {
    public static boolean isArmStrong(int n) {

        int nCopy = n;
        int sum = 0;
        while (n != 0) {
            int digit = n % 10;
            sum = sum + digit * digit * digit;
            n = n / 10;
        }
        if (sum == nCopy)
            return true;
        else
            return false;
    }

    public static void main(String[] args) {
        int n = 1000;
        for (int i = 1; i <= n; i++) {
            if (isArmStrong(i))
                System.out.println(i);
        }
    }
}

// Time Complexity - O(n log n)
//Space Complexity - O(1)
```



29) Lab29.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

```
//29. Find the Factorial of Given Number
```

```
public class Lab29 {  
  
    public static void main(String[] args) {  
        int n = 20;  
  
        long fact = 1;  
        for (int i = 1; i <= n; i++){  
            fact = fact * i;  
        }  
  
        System.out.println(fact);  
    }  
}
```

```
// Time Complexity - O(n)
```

```
//Space Complexity - O(1)
```

30) Lab30.java

```
package com.myjlc.loops;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
//30. Check whether given number is Strong Number or Not.
public class Lab30 {
    public static long factorial(int n) {
        if(n==0 || n==1)
            return 1;

        long fact = 1;
        for (int i = 1; i <= n; i++) {
            fact = fact * i;
        }
        return fact;
    }
    public static boolean isStrong(int n) {
        int nCopy = n;

        long sum = 0;
        while (n != 0) {
            int digit = n % 10;
            long fact = factorial(digit);
            sum += fact;
            n = n / 10;
        }
        return (sum == nCopy);
    }
    public static void main(String[] args) {
        int n = 100;
        boolean flag = isStrong(n);
        System.out.println(flag);
    }
}
// Time Complexity - O(log n)
//Space Complexity - O(1)
```



31) Lab31.java

```
package com.jlcindia;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
//31. Print the Strong Numbers from 1 to N
public class Lab31 {
    public static long factorial(int n) {
        long fact = 1;
        for (int i = 1; i <= n; i++) {
            fact = fact * i;
        }
        return fact;
    }
    public static boolean isStrong(int n) {
        int nCopy = n;

        long sum = 0;
        while (n != 0) {
            int digit = n % 10;
            long fact = factorial(digit);
            sum += fact;
            n = n / 10;
        }
        return (sum == nCopy);
    }
    public static void main(String[] args) {
        int n = 1000;

        for(int i=1;i<=n;i++) {
            if(isStrong(i))
                System.out.println(i);
        }
    }
}
// Time Complexity - O(n log n)
//Space Complexity - O(1)
```



32) Lab32.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//32. Print the Reverse of given Number

public class Lab32 {

    public static void main(String[] args) {
        int n = 12345;

        int rev = 0;

        while (n != 0) {
            rev = rev * 10 + n % 10;
            n = n / 10;
        }
        System.out.println(rev);
    }
}

// Time Complexity - O(d) where d- No of Digits
//Space Complexity - O(1)
```



33) Lab33.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//33. Check whether given number is Palindrome or Not

public class Lab33 {

    public static boolean isPalindrome(int n) {

        int nCopy=n;
        int rev = 0;

        while (n != 0) {
            rev = rev * 10 + n % 10;
            n = n / 10;
        }
        System.out.println(rev);
        return (rev==nCopy);
    }

    public static void main(String[] args) {
        int n = 121;

        boolean flag=isPalindrome(n);
        System.out.println(flag);

    }

}

// Time Complexity - O(d) where d- No of Digits
//Space Complexity - O(1)
```



34) Lab34.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

```
//34. Print the Digits of Given Number in same Order  
// n= 123 - 1, 2, 3
```

```
public class Lab34 {  
    public static void main(String[] args) {  
        int n = 12345;  
  
        int rev = 0;  
  
        while (n != 0) {  
            int digit = n % 10;  
            rev = rev * 10 + digit;  
            n = n / 10;  
        }  
  
        while (rev != 0) {  
            int digit = rev % 10;  
            System.out.println( digit % 10);  
            rev = rev / 10;  
        }  
    }  
}
```

```
// Time Complexity - O(d) where d- No of Digits
```

```
//Space Complexity - O(1)
```



35) Lab35.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

// 29) WAP to print the fibonocii series.
// 0 1 1 2 3 5 8 13

public class Lab35 {
    public static void main(String[] args) {

        int n =10;

        int a = 0;
        int b = 1;

        System.out.print(a+"\t"+b+"\t");

        for(int i=1;i<=n-2;i++) {
            int next = a+b;
            System.out.print(next+"\t");
            a=b;
            b=next;
        }
    }
}

// Time Complexity - O(n)
//Space Complexity - O(1)
```



36) Lab36.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//36. Print the following Series
//1,1,6,4,11,9,16,16,21,25

public class Lab36 {

    public static void main(String[] args) {

        int n = 5;

        int a = 1;
        int b=1;

        for (int i = 1; i <= n; i++) {
            //1. First Series
            System.out.print(a + "\t");
            a = a + 5;

            //2. Second Series
            b= i*i;
            System.out.print(b+"\t");
        }
    }

}

// Time Complexity - O(n)
//Space Complexity - O(1)
```



37) Lab37.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//37. Print the following Series
//0, 1, 3, 8, 6, 27, 9, 64, 12, 125

public class Lab37 {
    public static void main(String[] args) {

        int n = 5;

        int a = 0;
        int b=1;

        for (int i = 1; i <= n; i++) {
            //1. First Series
            System.out.print(a + "\t");
            a = a + 3;

            //2. Second Series
            b= i*i*i;
            System.out.print(b+"\t");
        }
    }
}

// Time Complexity - O(n)
//Space Complexity - O(1)
```



38) Lab38.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//38. Evaluate the following Services
// 1 + 2/x+3/x2+4/x3+....

public class Lab38 {
    public static double power(double x,double p) {
        double result=1;
        for(int i=1;i<=p;i++) {
            result= result*x;
        }
        return result;
    }

    public static void main(String[] args) {

        int n = 5;
        int x=2;

        double result = 0.0;
        for (int i = 1; i <= n; i++) {
            result = result + i/power(x,i-1);
        }
        System.out.println(result);
    }
}

// Time Complexity - O(n)/O(n*p)
//Space Complexity - O(1)
```



39) Lab39.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//39. Evaluate the following Services
// 1 - 2/x+3/x2-4/x3+....

public class Lab39 {
    public static double power(double x,double p) {
        double result=1;
        for(int i=1;i<=p;i++) {
            result= result*x;
        }
        return result;
    }
    public static void main(String[] args) {

        int n = 5;
        int x=2;
        double result = 0.0;
        int sign=1;
        for (int i = 1; i <= n; i++) {
            result = result + (sign)*( i/power(x,i-1));
            sign = sign*-1;
        }
        System.out.println(result);
    }
}

//Time Complexity - O(n)/O(n*p)
//Space Complexity - O(1)
```



40) Lab40.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
*/
```

```
//40. Evaluate the following Services
```

```
//  $1/(x)! + x/(x+1)! + x^2/(x+2)! + x^3/(x+3)! + \dots$ 
```

```
public class Lab40{
```

```
}
```

```
//Time Complexity -
```

```
//Space Complexity -
```



41) Lab41.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
*/
```

```
//41. Evaluate the following Services
```

```
//  $1/(x)! - x/(x+1)! + x^2/(x+2)! - x^3/(x+3)! + \dots$ 
```

```
public class Lab41{
```

```
}
```

```
//Time Complexity -
```

```
//Space Complexity -
```



42) Lab42.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
/*
42) Print the following Pattern
*****
*****
*****
*****
*****

*/
public class Lab42 {
    public static void main(String[] args) {

        int rows = 6;
        int stars = 6;

        for (int i = 1; i <= rows; i++) {
            // 1. Print Stars
            for (int k = 1; k <= stars; k++) {
                System.out.print(" * ");
            }
            // 2. Print New Line
            System.out.println("");
        }
    }
}

//Time Complexity - O(rows * stars)
//Space Complexity - O(1)
```



43) Lab43.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
/*
43) Print the following Pattern
    *
    **
   ***
  ****
 *****

 */

public class Lab43 {
    public static void main(String[] args) {

        int rows = 6;
        int stars = 1;

        for (int i = 1; i <= rows; i++) {
            // 1. Print Stars
            for (int k = 1; k <= stars; k++) {
                System.out.print(" * ");
            }
            // 2. Print New Line
            System.out.println("");

            stars=stars+1;
        }
    }
}

//Time Complexity - O(rows * stars)
//Space Complexity - O(1)
```



44) Lab44.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
*/
```

```
/*
```

44) Print the following Pattern

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

```
*/
```

```
public class Lab44 {
    public static void main(String[] args) {

        int rows = 7;
        int stars = 1;

        for (int i = 1; i <= rows/2+1; i++) {
            // 1. Print Stars
            for (int k = 1; k <= stars; k++) {
                System.out.print(" * ");
            }
            // 2. Print New Line
            System.out.println("");

            stars=stars+1;
        }
    }
}
```



```
        for (int i = 1; i <= rows/2; i++) {

            // 1. Print Stars
            for (int k = 1; k <= stars-2; k++) {
                System.out.print(" * ");
            }
            // 2. Print New Line
            System.out.println("");

            stars=stars-1;

        }

    }

}

//Time Complexity - O(rows * stars)
//Space Complexity - O(1)
```

45) Lab45.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

/*
45) Print the following Pattern
    *
    * * *
    * * * * *
    * * * * * *
    */
```



```
public class Lab45 {  
    public static void main(String[] args) {  
  
        int rows = 10;  
        int spaces=25;  
        int stars = 1;  
  
        for (int i = 1; i <= rows; i++) {  
            //1.Print Spaces  
            for (int k = 1; k <= spaces; k++) {  
                System.out.print(" ");  
            }  
  
            // 2. Print Stars  
            for (int k = 1; k <= stars; k++) {  
                System.out.print(" * ");  
            }  
  
            // 3. Print New Line  
            System.out.println("");  
  
            spaces = spaces - 2;  
            stars=stars+2;  
        }  
    }  
}  
  
//Time Complexity - O(rows * stars)  
//Space Complexity - O(1)
```



46) Lab46.java

```
package com.jlcindia;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
/*
46) Print the following Pattern
    *
   ***
  *****
 *****
 *****
   ***
    *
*/
public class Lab46 {
    public static void main(String[] args) {
        int rows = 11;
        int spaces=25;
        int stars = 1;

        for (int i = 1; i <= rows/2; i++) {
            //1.Print Spaces
            for (int k = 1; k <= spaces; k++) {
                System.out.print(" ");
            }

            // 2. Print Stars
            for (int k = 1; k <= stars; k++) {
                System.out.print(" * ");
            }

            // 3. Print New Line
            System.out.println("");

            spaces = spaces - 2;
            stars=stars+2;
        }
    }
}
```



```
        for (int i = 1; i <= rows/2+1; i++) {  
            //1.Print Spaces  
            for (int k = 1; k <= spaces; k++) {  
                System.out.print(" ");  
            }  
  
            // 2. Print Stars  
            for (int k = 1; k <= stars; k++) {  
                System.out.print(" * ");  
            }  
  
            // 3. Print New Line  
            System.out.println("");  
  
            spaces = spaces + 2;  
            stars=stars-2;  
        }  
    }  
}  
  
//Time Complexity - O(rows * stars)  
//Space Complexity - O(1)
```



LeetCode Math Problems

1	LC-7 – Reverse Integer
2	LC-258 – Add digits
3	LC-9 – Palindrome Number
4	LC-866 – Prime Palindrome
5	LC-507 – Perfect Number
6	LC-1134 – ArmStrong Number
7	LC-326 – Power of 3
8	LC-342 – Power of 4
9	LC-50 – Power(x,n)



LeetCode7.java

//7. Reverse Integer

```
public class LeetCode7 {  
  
    public static int reverse(int num) {  
        long rev=0;  
        while(num!=0) {  
            rev = rev*10+num%10;  
            num=num/10;  
        }  
  
        if(rev<Integer.MIN_VALUE || rev>Integer.MAX_VALUE)  
            rev=0;  
  
        return (int)rev;  
    }  
  
    public static void main(String[] args) {  
        int num=1534236469;  
        int result = reverse(num);  
        System.out.println(result);  
    }  
}
```



LeetCode258.java

//258. Add Digits

```
public class LeetCode258 {  
  
    public static int addDigits(int num) {  
        int sum = 10;  
        while (sum >= 10) {  
            sum = 0;  
            while (num != 0) {  
                sum = sum + num % 10;  
                num = num / 10;  
            }  
            System.out.println(sum);  
            num = sum;  
        }  
        return sum;  
    }  
  
    public static void main(String[] args) {  
        int num = 123456;  
        int result = addDigits(num);  
        System.out.println("Result : "+result);  
    }  
}
```



LeetCode9.java

//9. Palindrome Number

```
public class LeetCode9 {  
  
    public static boolean isPalindrome(int num) {  
        if (num < 0)  
            return false;  
  
        int ncopy = num;  
  
        int rev = 0;  
        while (num != 0) {  
            rev = rev * 10 + num % 10;  
            num = num / 10;  
        }  
  
        return (ncopy == rev);  
    }  
  
    public static void main(String[] args) {  
        int num = 1232;  
        boolean result = isPalindrome(num);  
        System.out.println(result);  
    }  
}
```



LeetCode866.java

//866. Prime Palindrome

```
public class LeetCode866 {

    public static boolean isPalindrome(int num) {
        if (num < 0)
            return false;

        int ncopy = num;

        int rev = 0;
        while (num != 0) {
            rev = rev * 10 + num % 10;
            num = num / 10;
        }

        return (ncopy == rev);
    }

    public static boolean isPrime(int num) {

        if (num < 2)
            return false;

        for (int i = 2; i <= Math.sqrt(num); i++) {
            if (num % i == 0)
                return false;
        }

        return true;
    }
}
```



```
public static int primePalindrome(int num) {  
  
    if (isPrime(num) && isPalindrome(num))  
        return num;  
  
    while (true) {  
        if (isPrime(num) && isPalindrome(num))  
            return num;  
        num++;  
        if (10_000_000 < num && num < 100_000_000)  
            num = 100_000_000;  
    }  
}  
  
public static void main(String[] args) {  
    int num = 13;  
    int result = primePalindrome(num);  
    System.out.println(result);  
}  
}
```



LeetCode507.java

//507. Perfect Number

```
public class LeetCode507 {  
    public static boolean checkPerfectNumber(int num) {  
  
        if (num % 2 != 0) {  
            return false;  
        }  
  
        int sum = 0;  
        for (int i = 1; i <= num / 2; i++) {  
            if (num % i == 0)  
                sum += i;  
        }  
  
        return (sum == num);  
    }  
  
    public static void main(String[] args) {  
        int num = 28;  
        boolean result = checkPerfectNumber(num);  
        System.out.println(result);  
    }  
}
```



LeetCode1134.java

//1134 - Armstrong Number

```
public class LeetCode1134 {  
  
    public static boolean isArmstrong(int num) {  
        int k = (num + "").length();  
        int sum = 0;  
        int numCopy=num;  
        while(num>0) {  
            int digit = num%10;  
            sum = sum + (int)Math.pow(digit, k);  
            num = num/10;  
        }  
  
        return (sum == numCopy);  
    }  
  
    public static void main(String[] args) {  
        boolean result = isArmstrong(153);  
        System.out.println(result);  
    }  
}
```



LeetCode326.java

//326. Power of Three

```
public class LeetCode326 {  
    public static boolean isPowerOfThree(int num) {  
        if (num == 0) return false;  
  
        while (num % 3 == 0) {  
            num = num / 3;  
        }  
        return (num == 1);  
    }  
}
```

LeetCode326R.java

//326. Power of Three

```
public class LeetCode326R {  
    public static boolean isPowerOfThree(int num) {  
        if (num == 0)  
            return false;  
  
        if (num == 1)  
            return true;  
  
        if (num % 3 != 0)  
            return false;  
  
        return isPowerOfThree(num / 3);  
    }  
}
```



LeetCode342.java

//342. Power of Four

```
public class LeetCode342 {  
    public static boolean isPowerOfFour(int num) {  
        if (num == 0) {  
            return false;  
        }  
        while (num % 4 == 0) {  
            num = num / 4;  
        }  
        return (num == 1);  
    }  
}
```

LeetCode342R.java

//342. Power of Four

```
public class LeetCode342R{  
    public static boolean isPowerOfFour(int num) {  
  
        if (num == 0) return false;  
  
        if (num == 1) return true;  
  
        if (num % 4 != 0)  
            return false;  
  
        return isPowerOfFour(num / 4);  
    }  
}
```



LeetCode50.java

//50. Pow(x, n)

```
public class LeetCode50 {  
    public static double power(double x, long n) {  
  
        if (n == 0)  
            return 1;  
  
        if (n == 1)  
            return x;  
  
        if (n < 0)  
            return power(1 / x, -n);  
  
        double result = power(x * x, n / 2);  
  
        if (n % 2 == 1)  
            result = result * x;  
  
        return result;  
    }  
  
    public static double myPow(double x, int n) {  
        return power(x, n);  
    }  
}
```